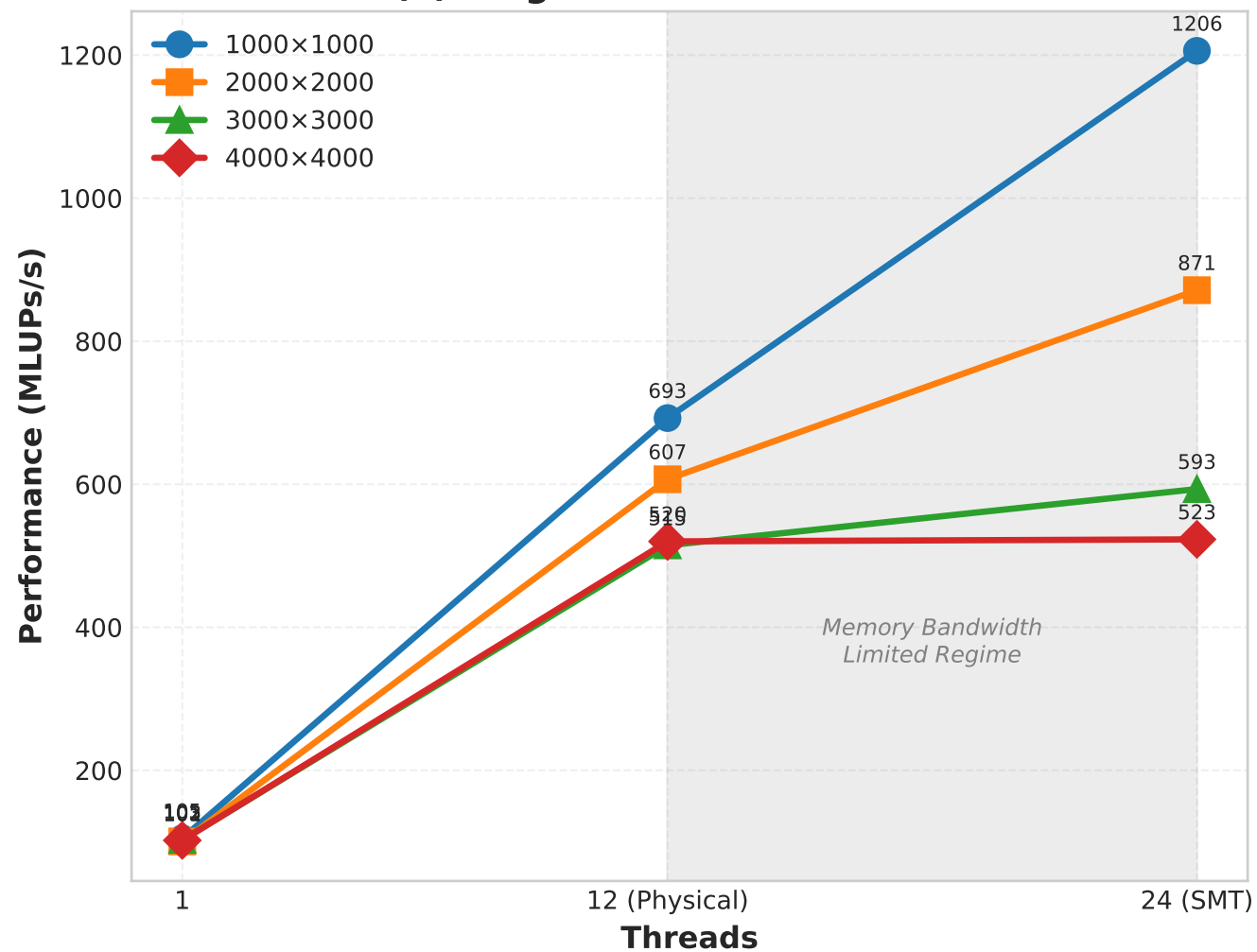


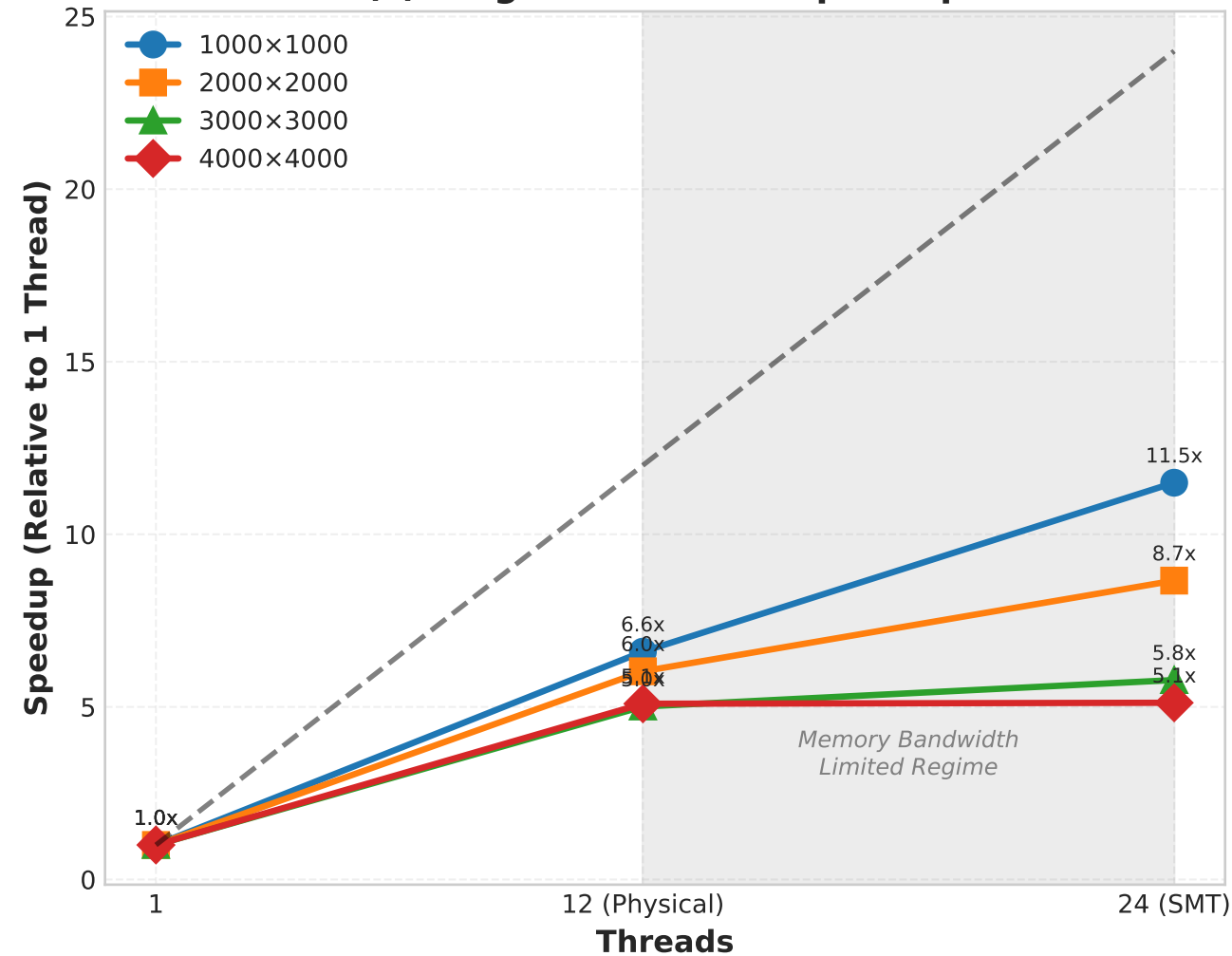
Thread Scaling Analysis: SMT Efficiency and Memory Bandwidth Limitations

2000 Time Steps

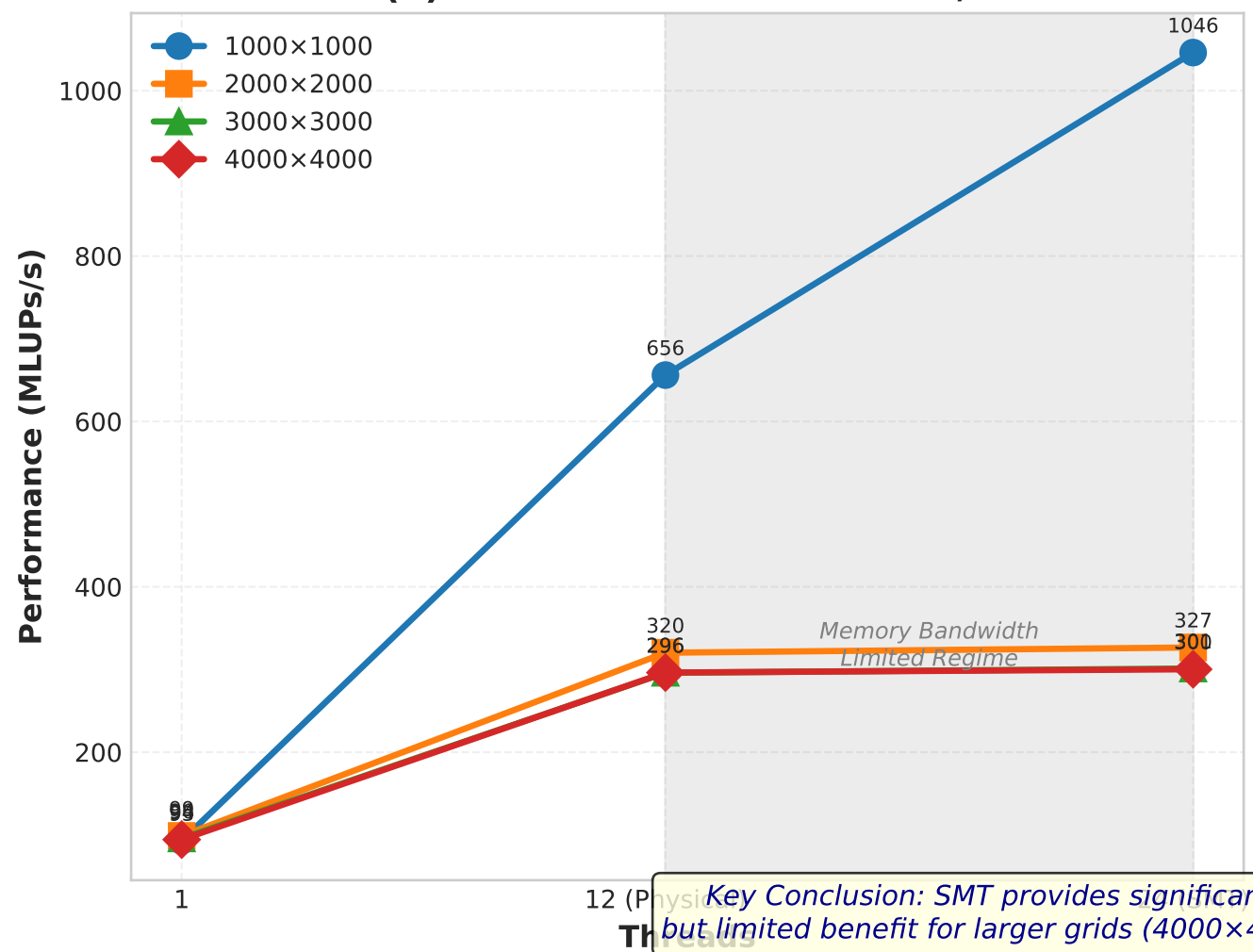
(a) Single Precision - MLUPs/s



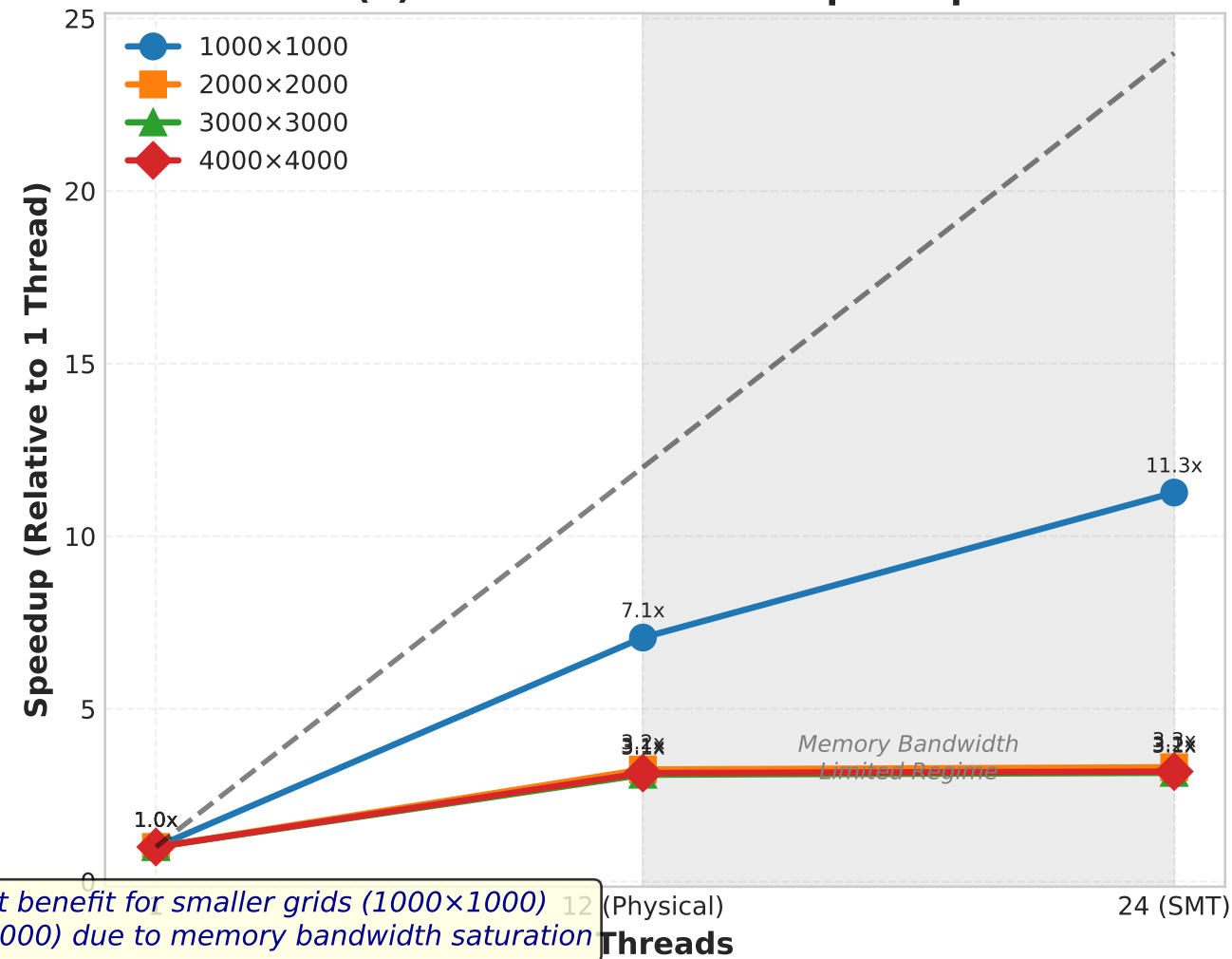
(c) Single Precision - Speedup



(b) Double Precision - MLUPs/s



(d) Double Precision - Speedup



Key Conclusion: SMT provides significant benefit for smaller grids (1000×1000) but limited benefit for larger grids (4000×4000) due to memory bandwidth saturation.

Hardware: AMD Ryzen 9 3900XT (12 Cores / 24 Threads) | Compiler: NVIDIA HPC SDK 24.5 (nvfortran) | OS: Linux